C

STM Exercise 8

Phase Diagrams

(Fe-C)

IRON-CARBON (Fe-C) PHASE DIAGRAM

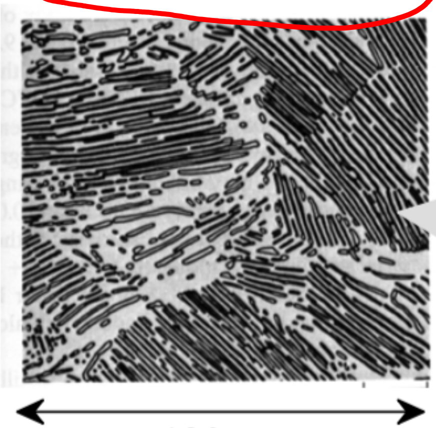
- 2 important points

  -Eutectic (A):

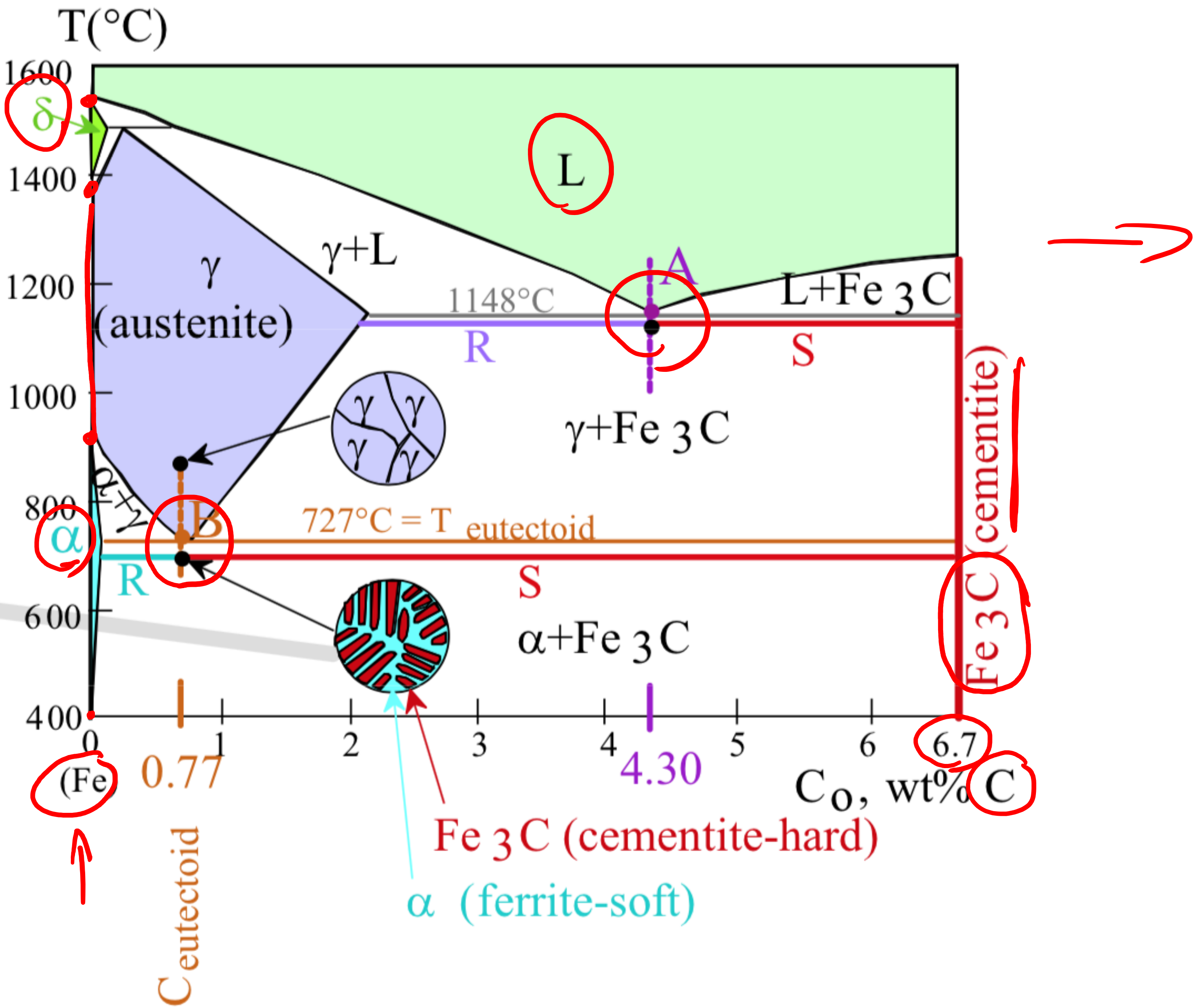
  $L \Rightarrow \gamma + Fe_3C$

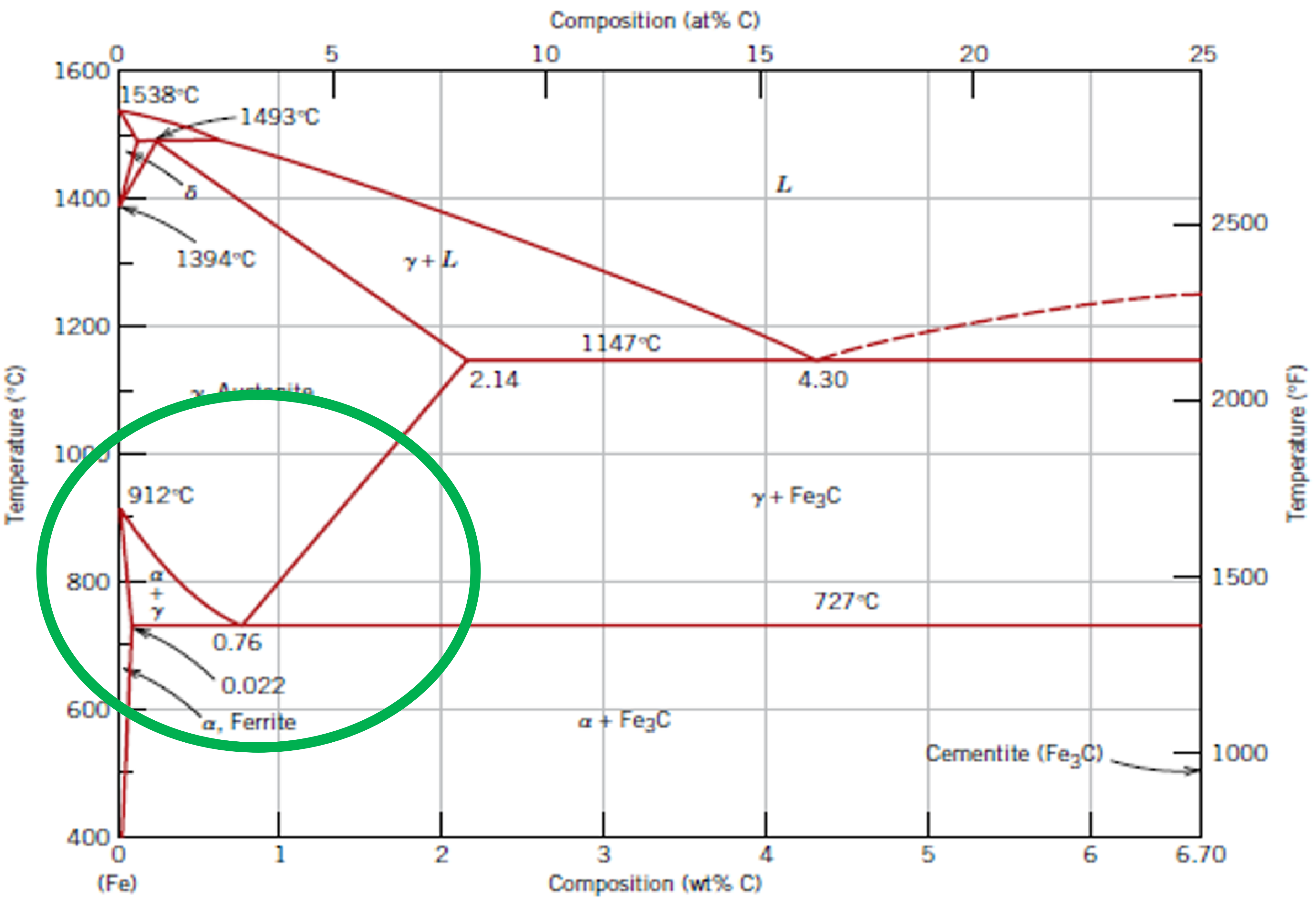
  -Eutectoid (B):

  $\gamma \Rightarrow \alpha + Fe_3C$

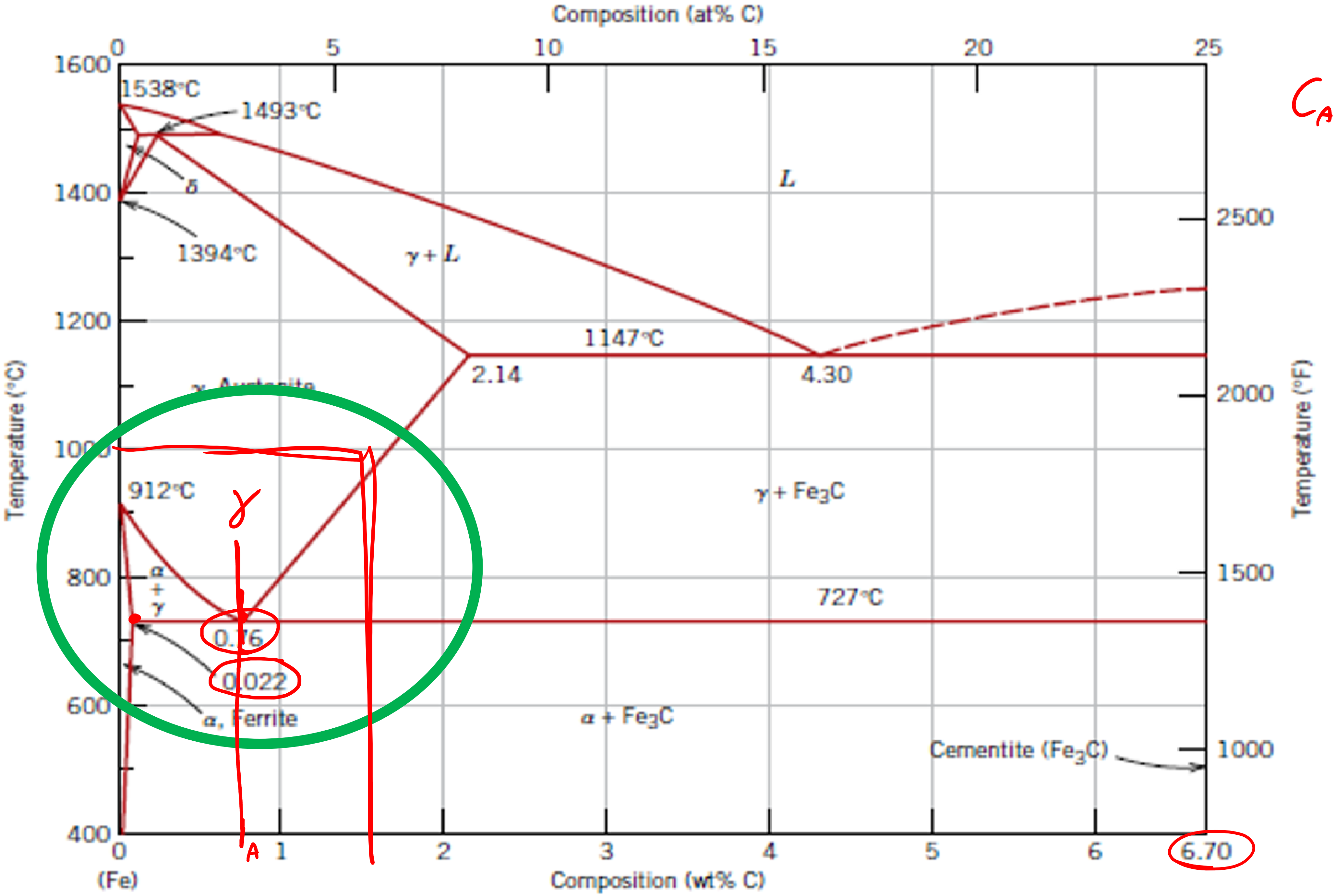


Result: Pearlite = alternating layers of α and Fe_3C phases.



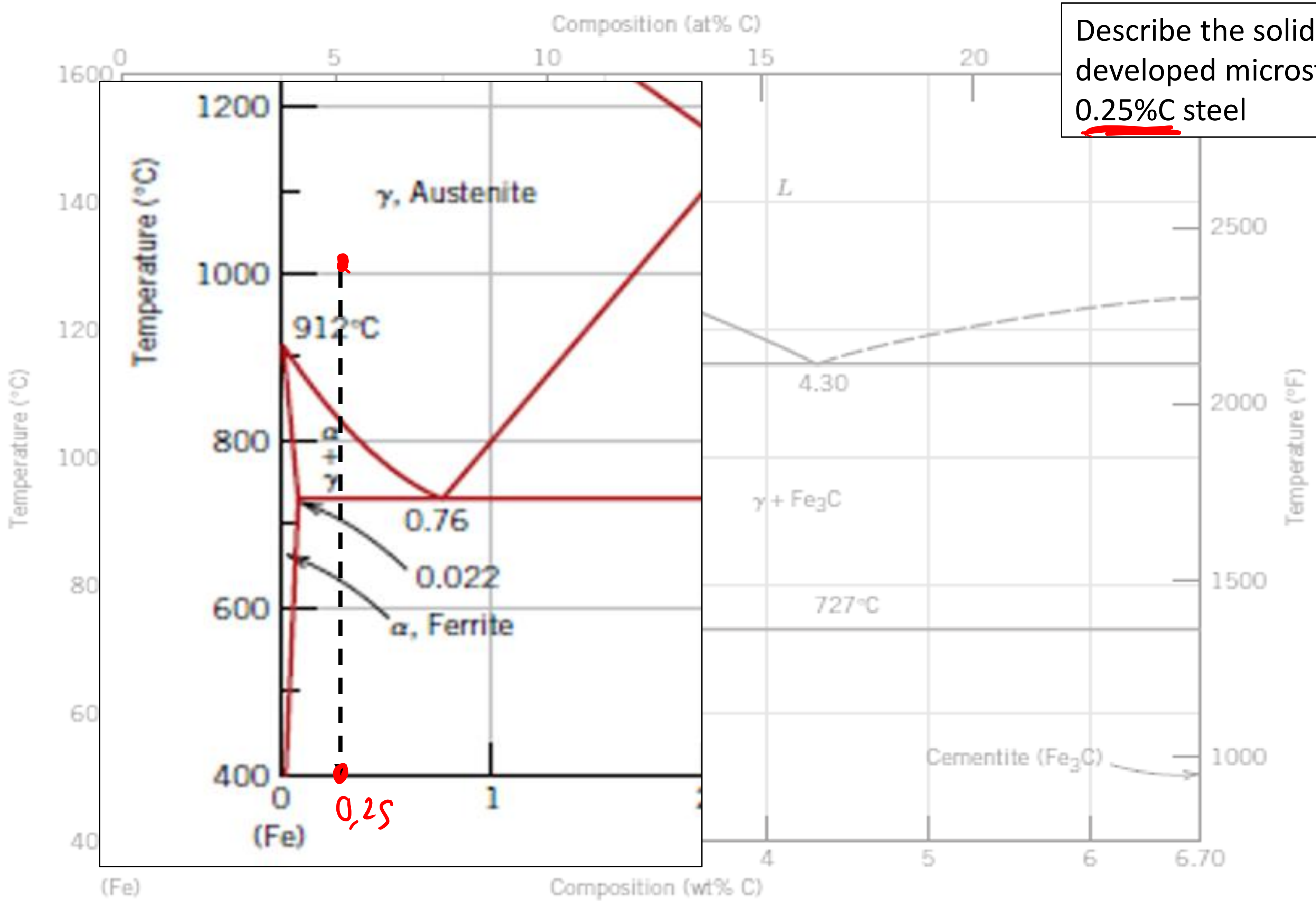


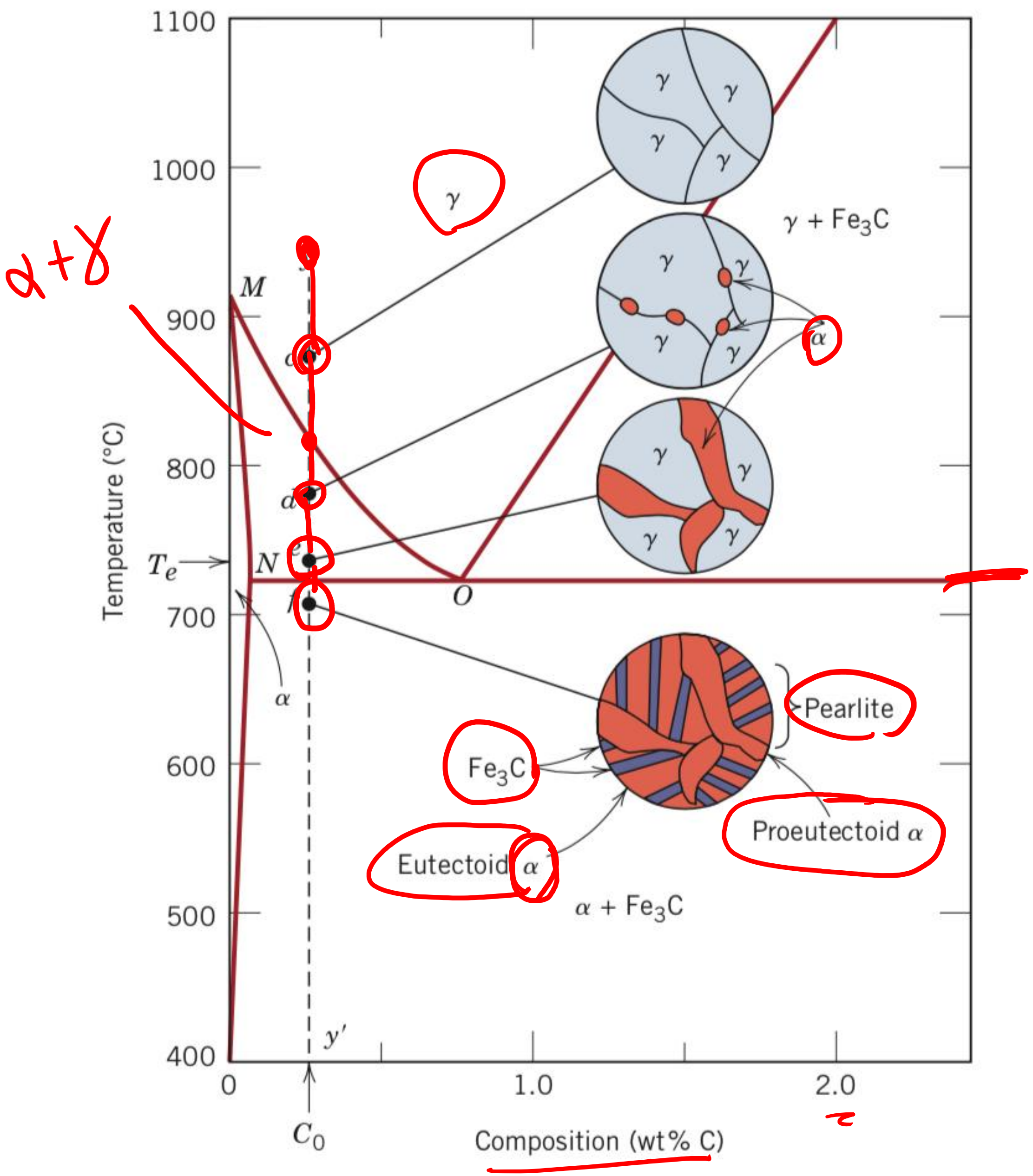
Composition (at% C)
0
5
10
15
20
25
1600
1538°C
1493°C
δ
1400
1394°C
L
2500
γ + L
1200
1147°C
2.14
4.30
2000
1000
912°C
γ + Fe3C
800
α
+
γ
727°C
1500
0.76
0.022
600
α, Ferrite
α + Fe3C
Cementite (Fe3C)
1000
400
0
(Fe)
1
2
3
4
5
6
6.70
Composition (wt% C)
Temperature (°C)
Temperature (°F)



$C_A \begin{cases} 0{,}76\%\ C \\ \cancel{99{,}24\%\ Fe} \end{cases}$

Describe the solidification (evidencing developed microstructures) of a 0.25%C steel





PRIMARY

α Proeutectoid alpha(ferrite)

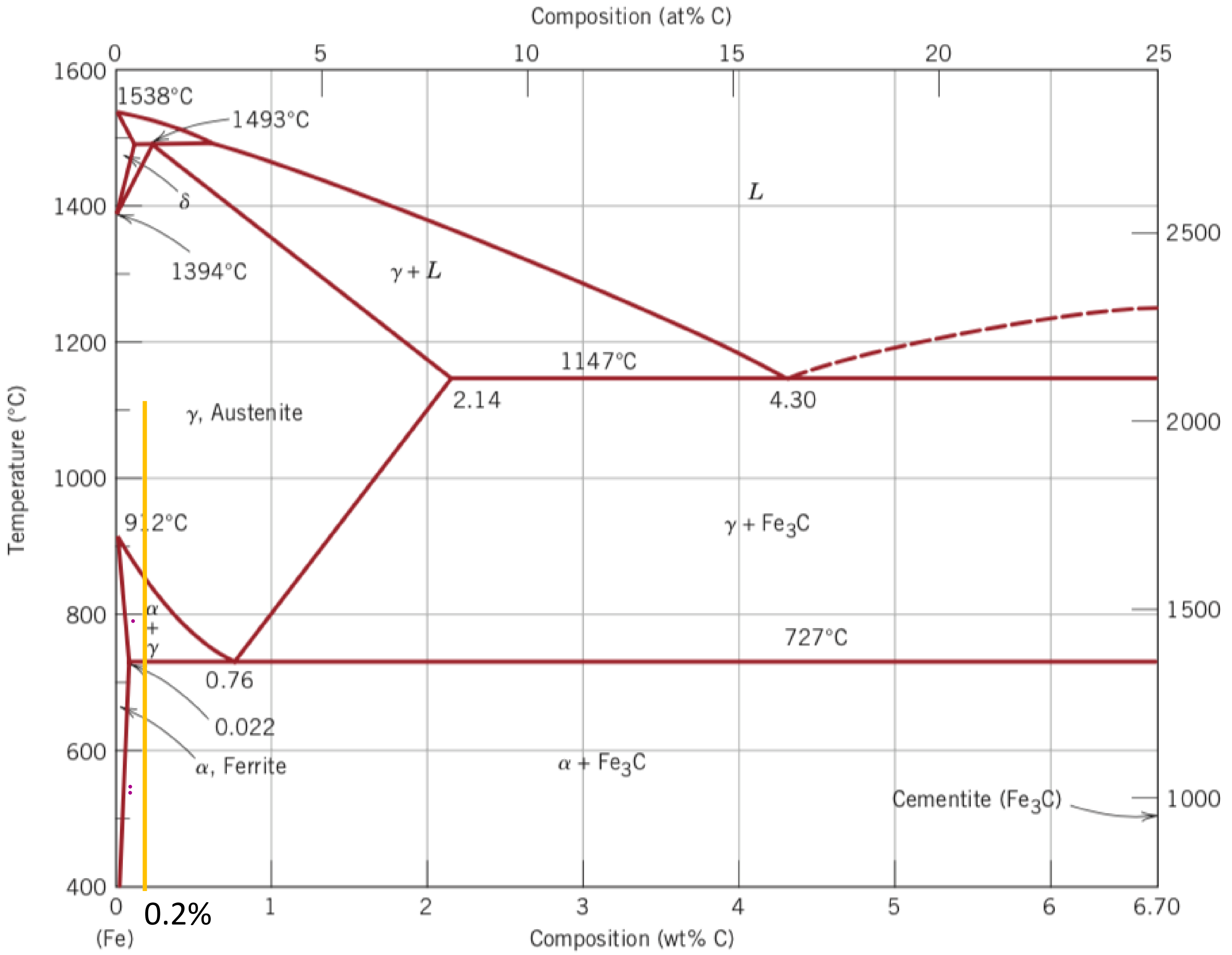
Eutectoid reaction: Austenite -> Ferrite + Cementite

γ α Fe_3C

Pearlite

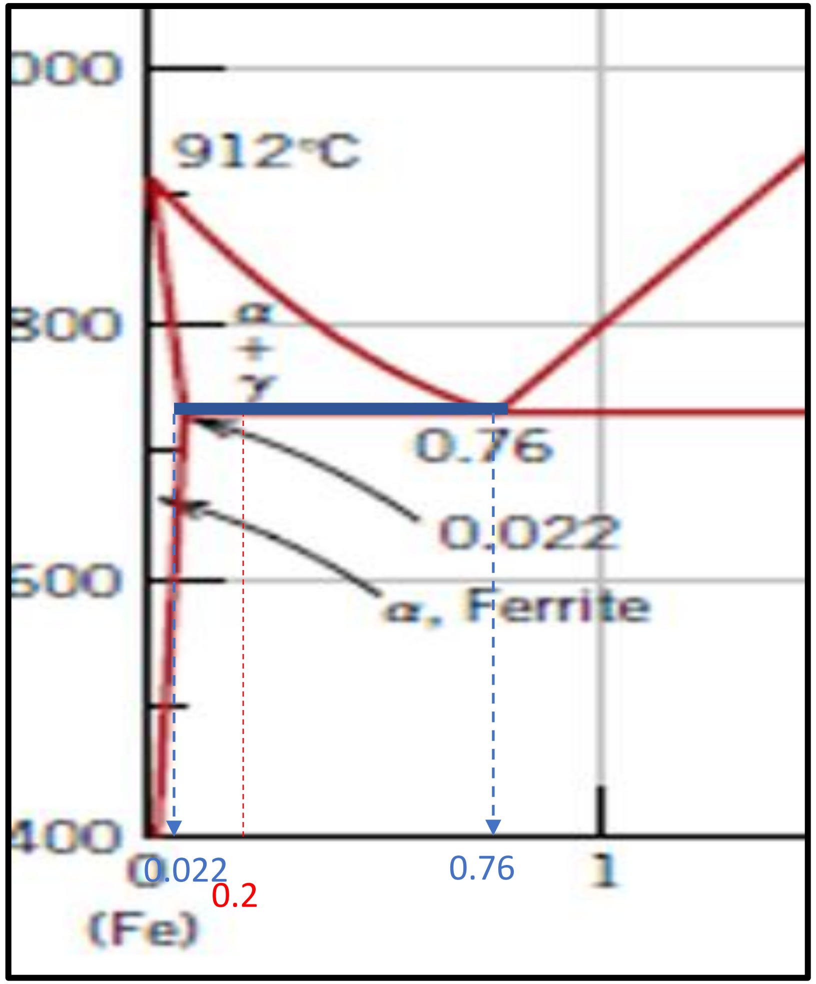
Use the Fe-Fe_3C phase diagram to calculate:

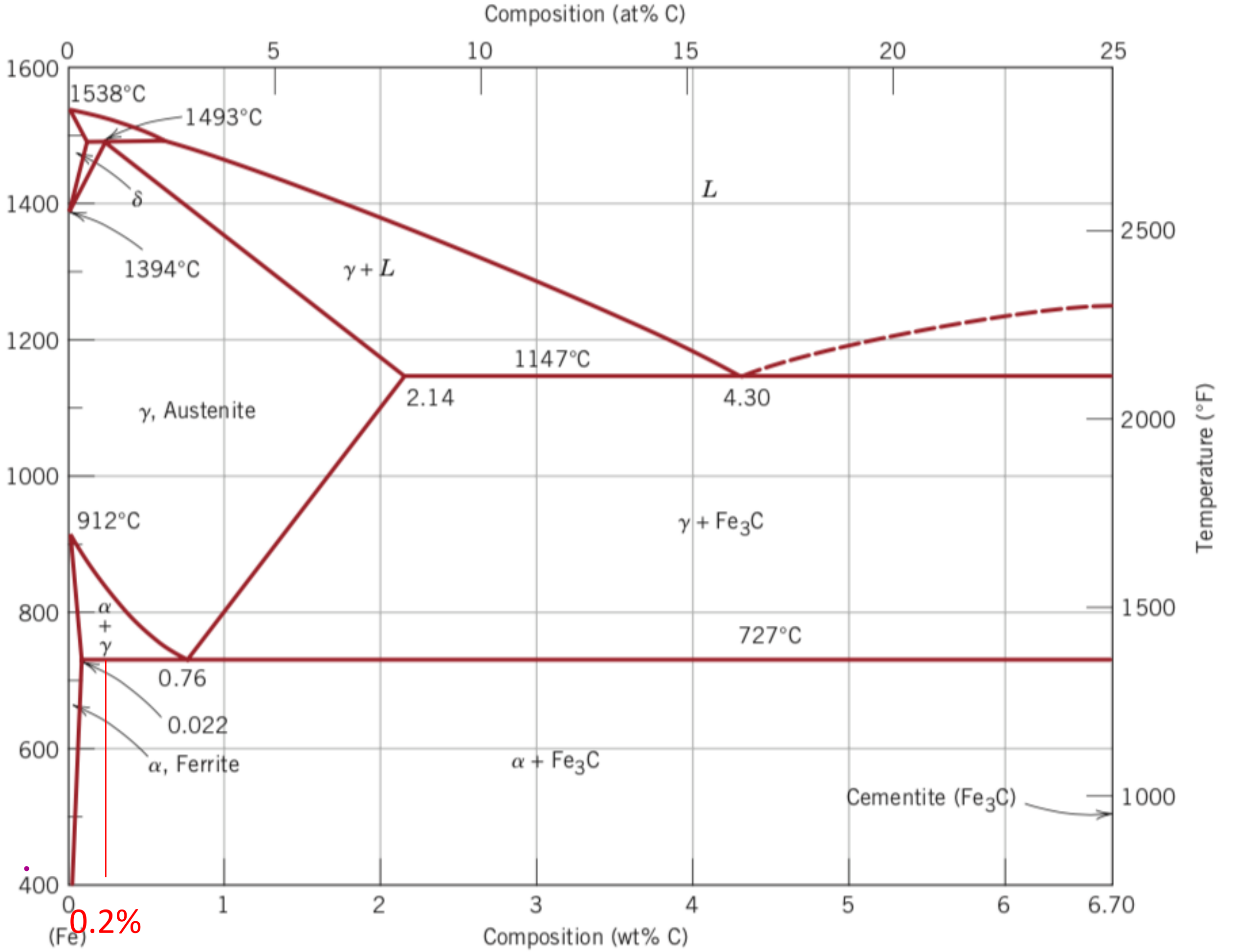
-the percentage of primary (pro-eutectoid) ferrite for a 0,2 % C steel



Use the Fe-Fe_3C phase diagram to calculate:

-the percentage of primary (pro-eutectoid) ferrite for a 0,2 % C steel



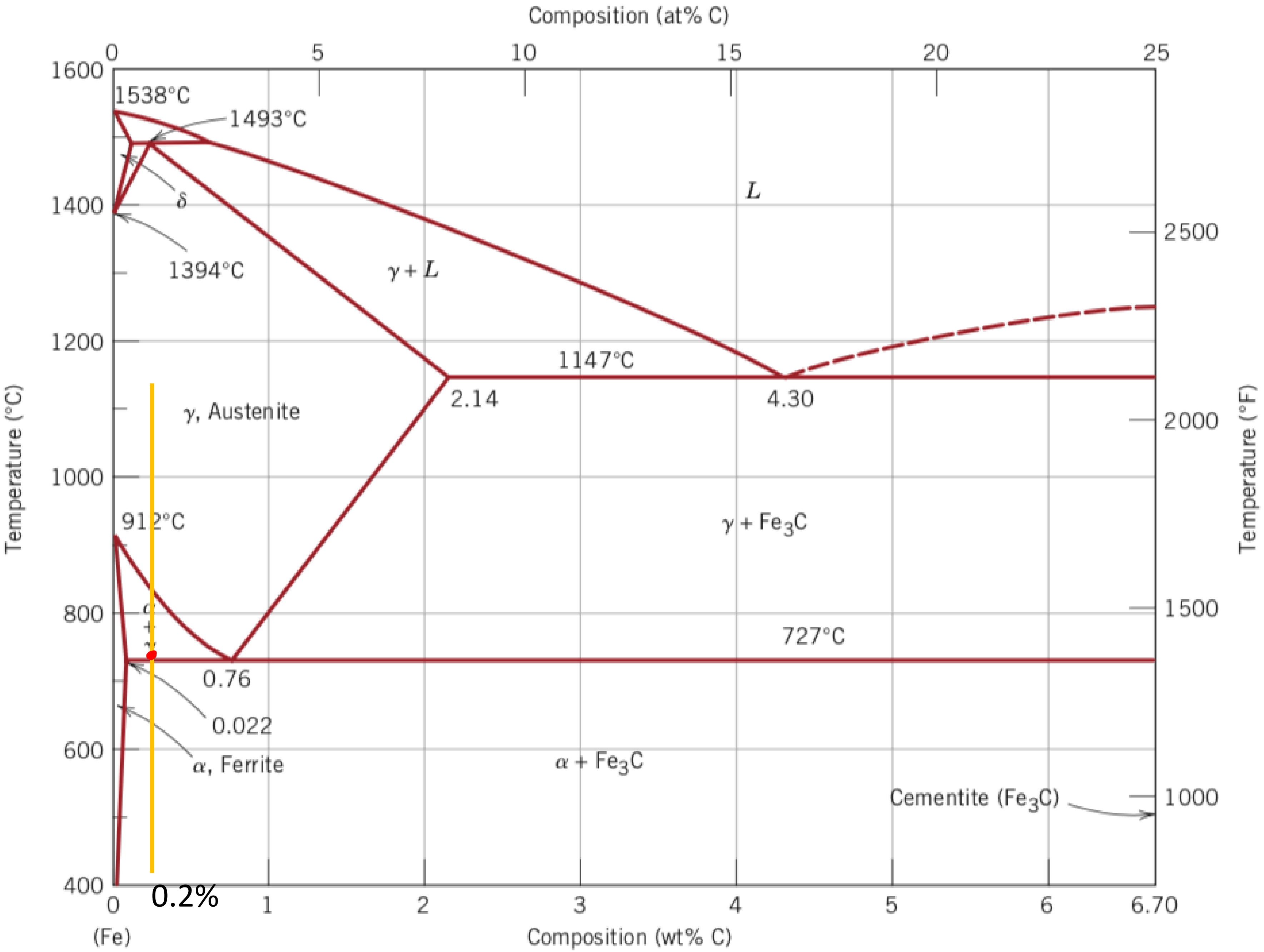


Lever rule on the eutectoidic tie line (blue line in figure):

$\%\alpha = (0.76-0.2)/(0.76-0.022)*100 = 76\%$

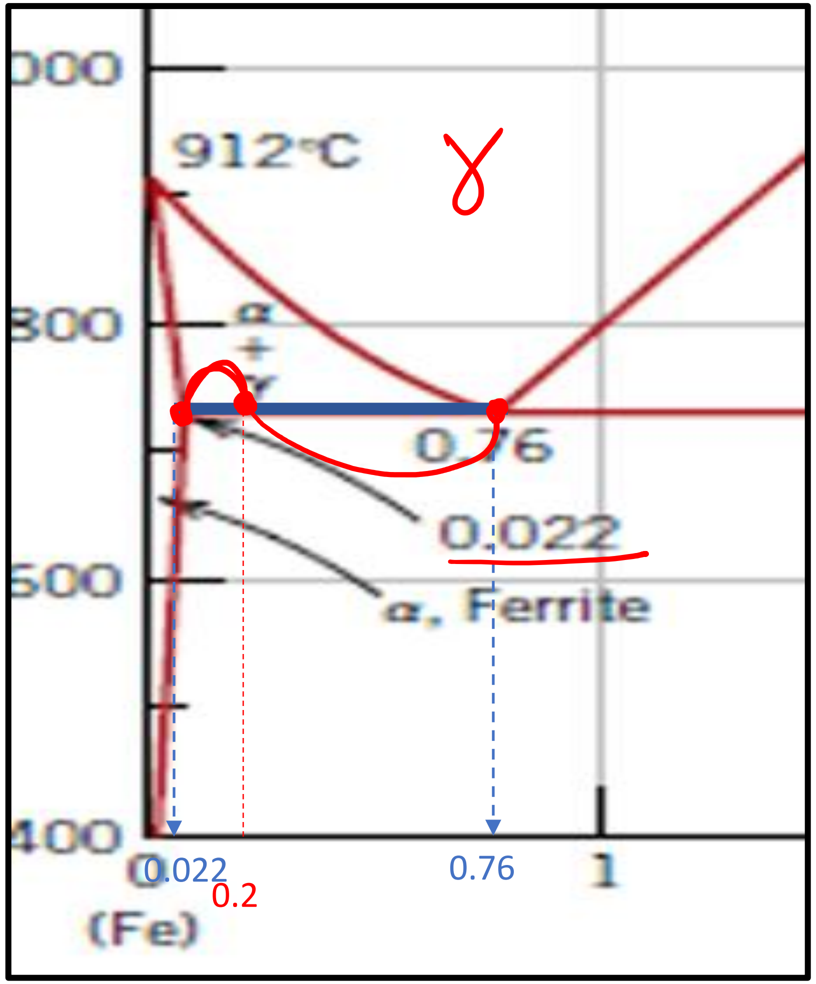
Use the Fe-Fe3C phase diagram to calculate:

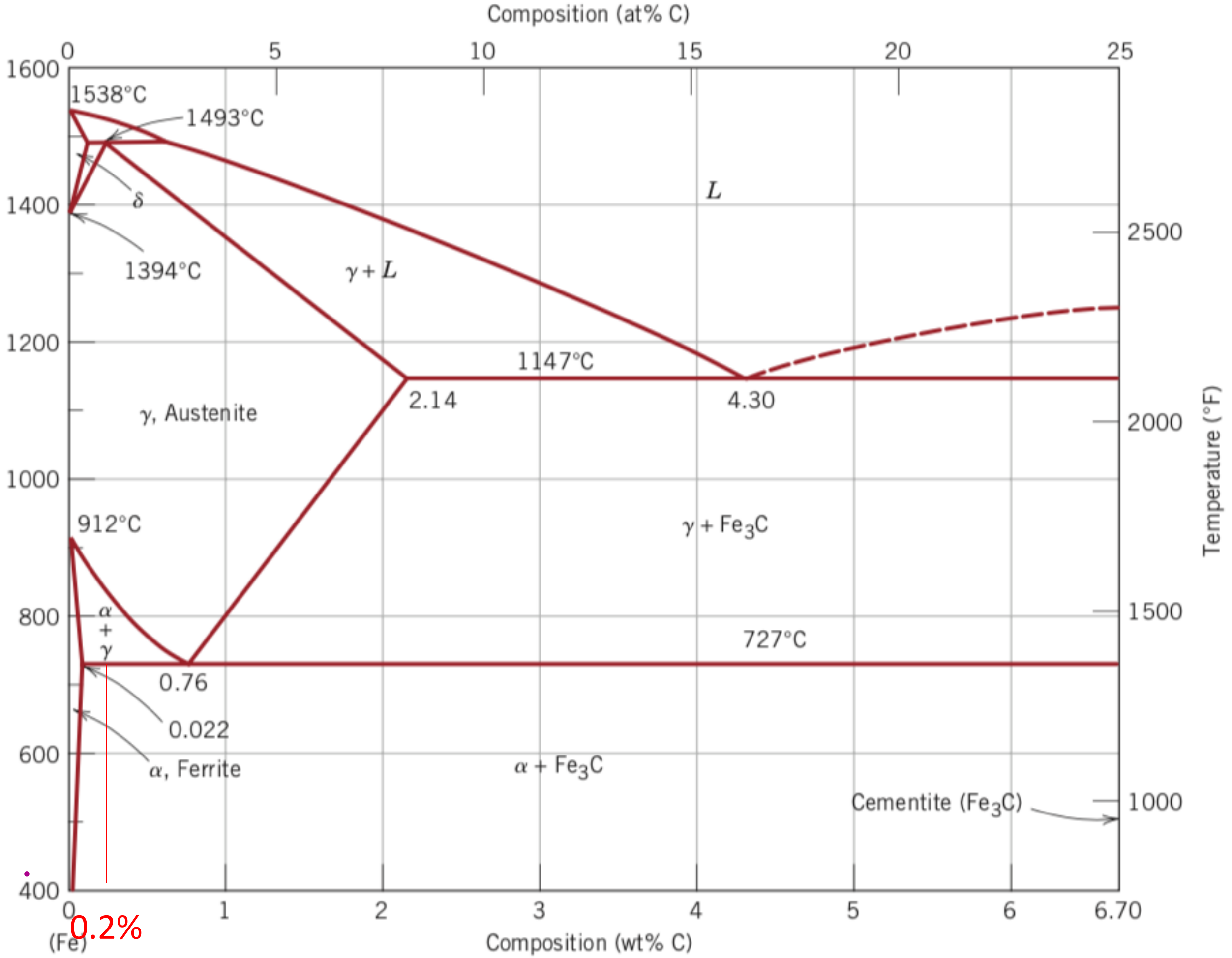
-the percentage of pearlite for a 0,2 % C steel



Use the Fe-Fe_3C phase diagram to calculate:

-the percentage of pearlite for a 0,2 % C steel





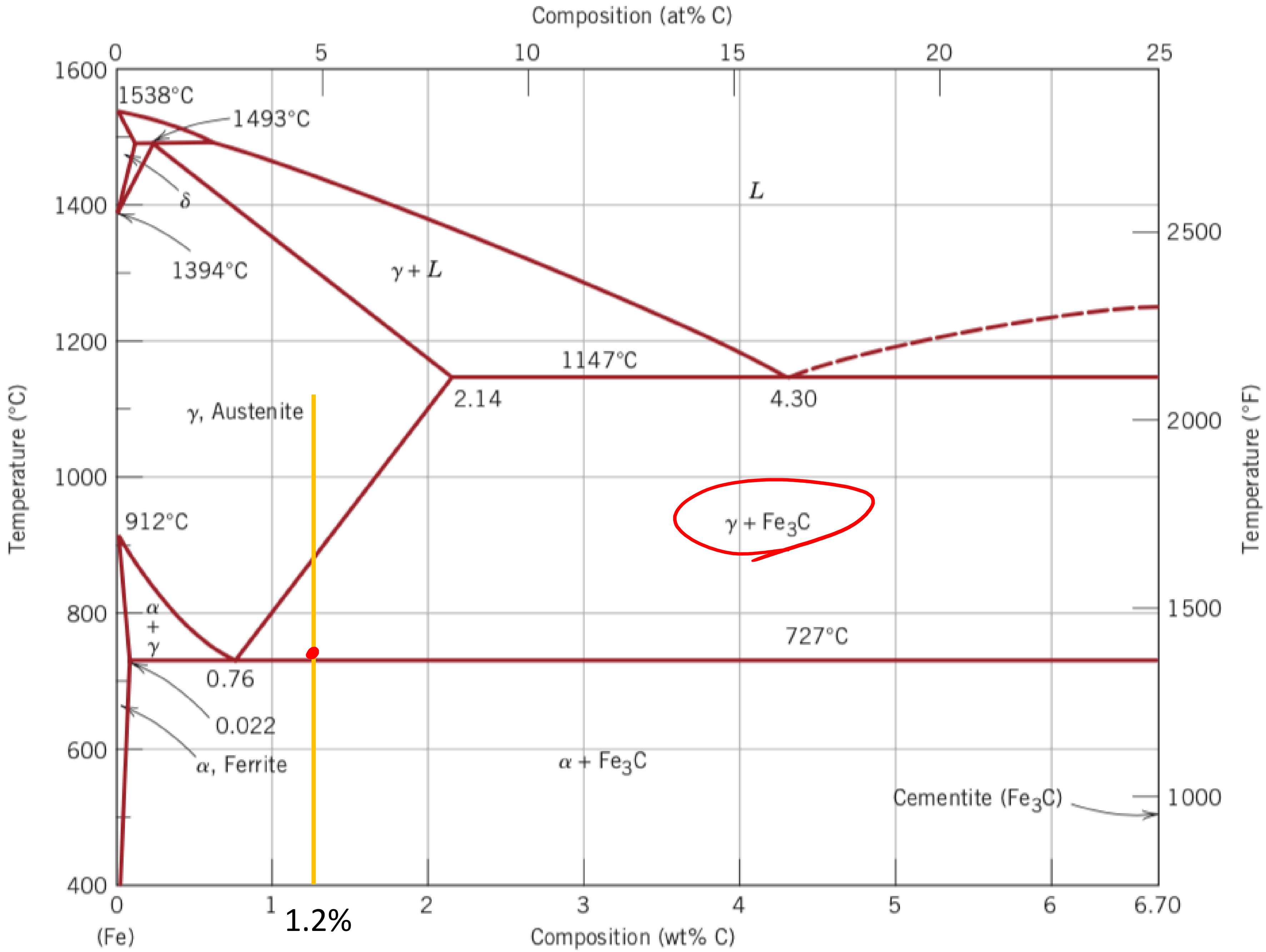
Lever rule on the eutectoidic tie line (blue line in figure):

$\%\gamma = (0.2-0.022)/(0.76-0.022)*100 = 24\%$

The % of pearlite is equal to the one of austenite just before the eutectoid

Use the Fe-Fe3C phase diagram to calculate:

-the percentage of pearlite for a 1,2 % C steel

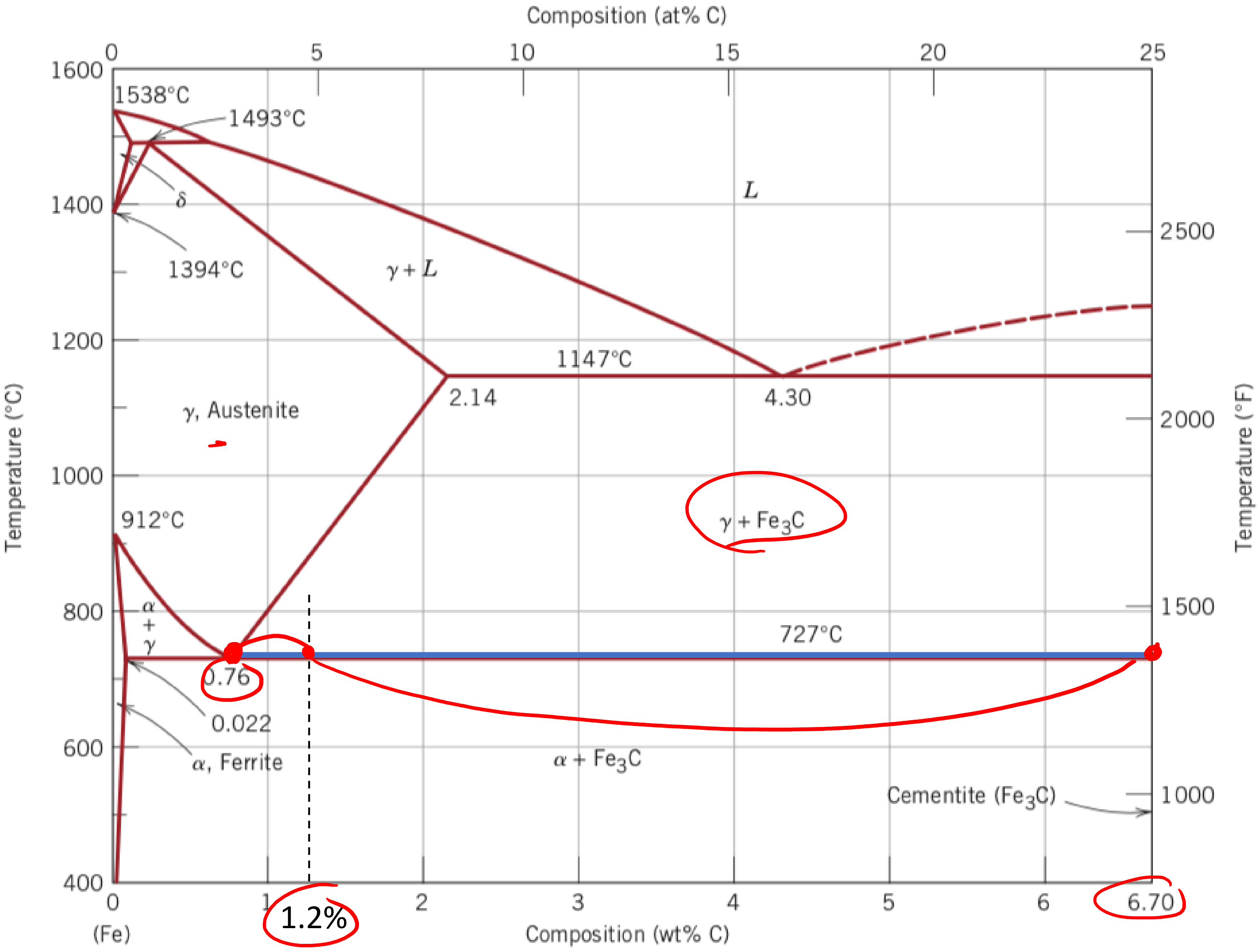


Use the Fe-Fe3C phase diagram to calculate:

-the percentage of pearlite for a 1,2 % C steel

During eutectoidic transformation all the austenite (γ) is transformed in α+Fe3C pearlitic structure.
So, the % of pearlite is equal to the % of γ at the beginning of the eutectoidic transformation
% γ (723°C)= (6.70-1.2)/(6.70-0.76)*100=~~92.5~~%

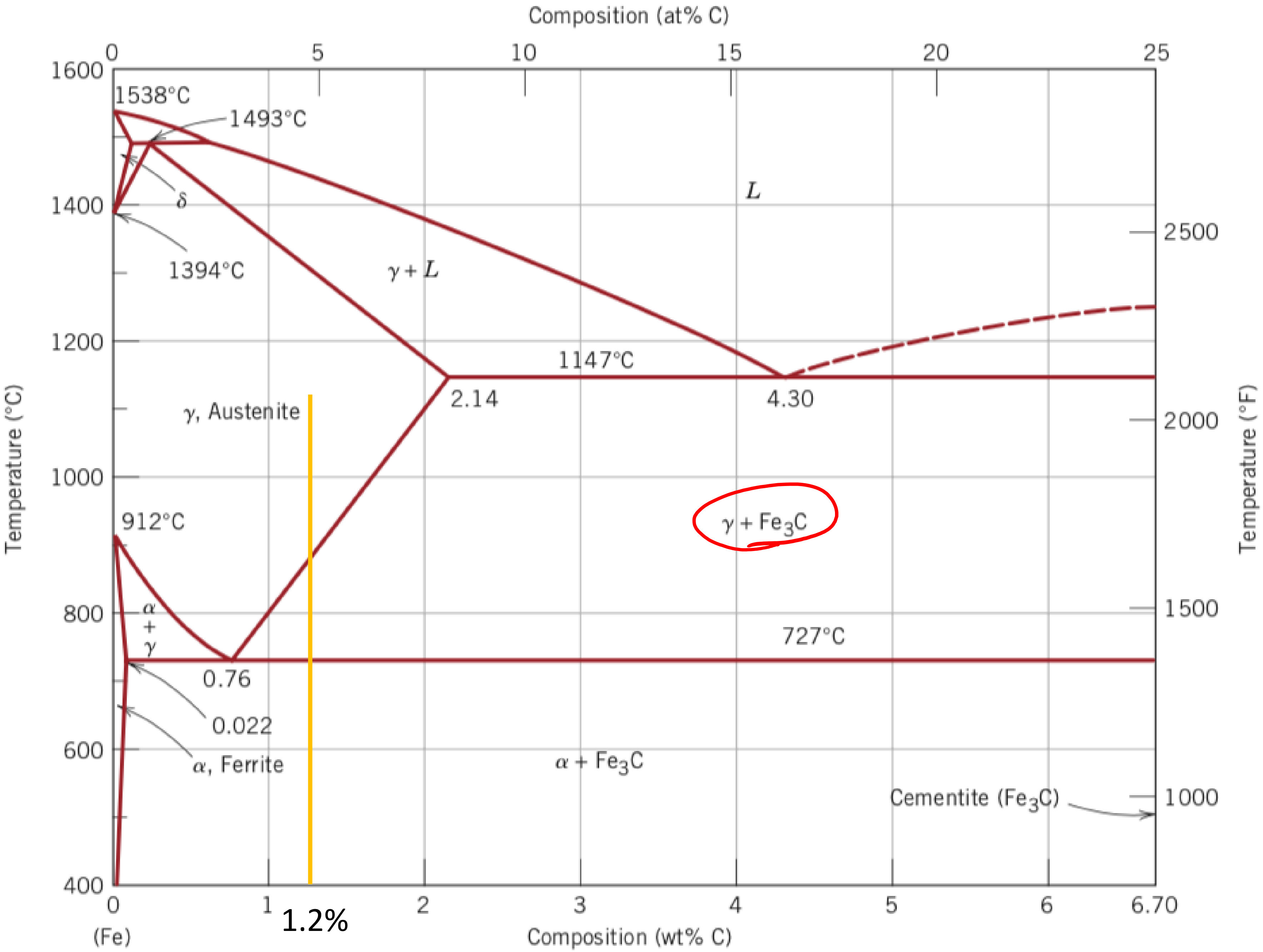
92,6%



Use the Fe-Fe3C phase diagram to calculate:

-the percentage of primary (pro-eutectoid) cementite for a 1,2 % C steel

$(Fe_3C)_p =$



Use the Fe-Fe3C phase diagram to calculate:

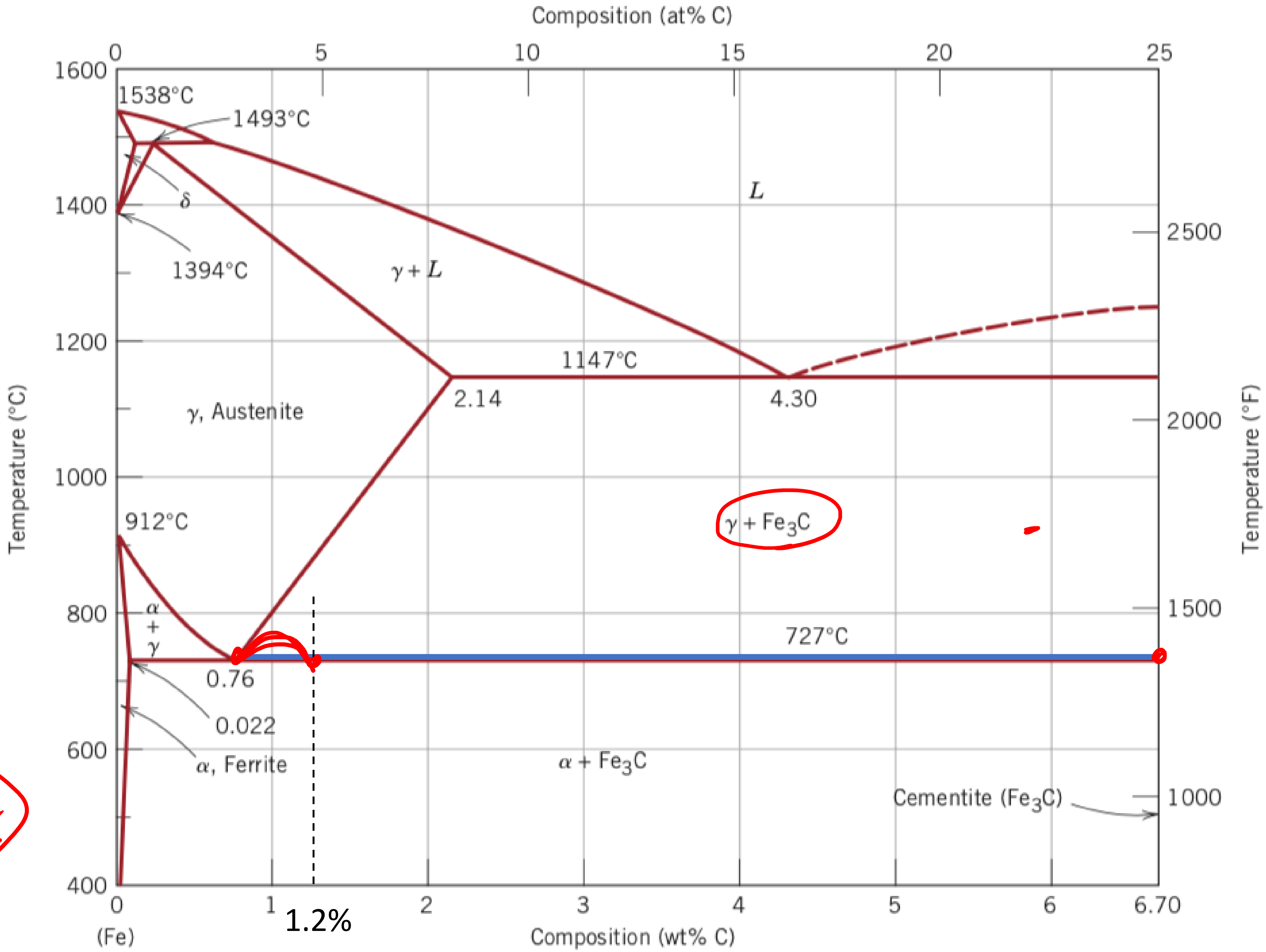
-the percentage of primary (pro-eutectoid) cementite for a 1,2 % C steel

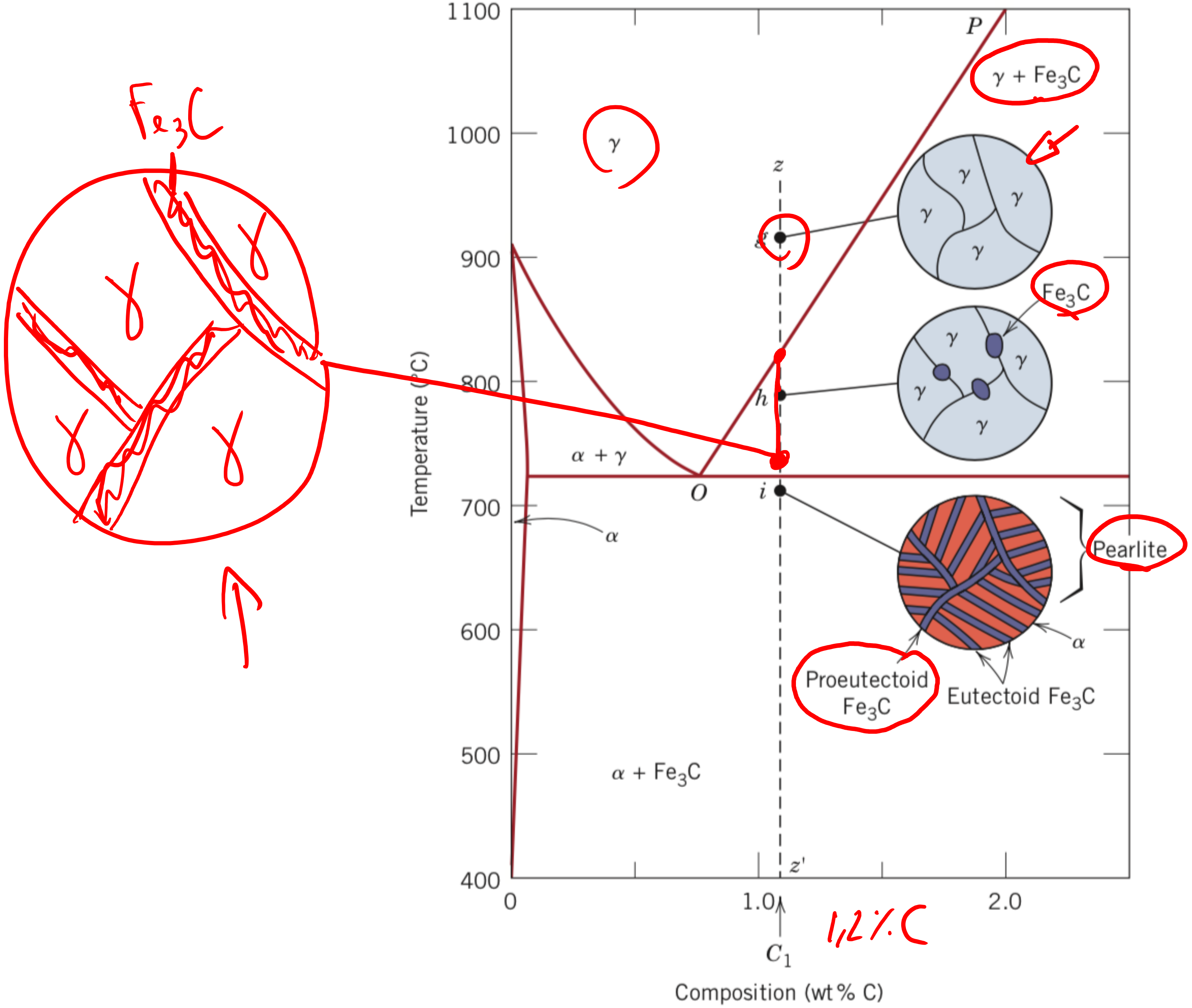
Lever rule at 727°C
%Fe3C (primary)= (1.2-0.76)/(6.70-0.76)*100= 7.4%

7,4%

100% - PEARLITE = $(Fe_3C)_p$

$(Fe_3C)_p = 100 - 92,6 = 7,4\%$





$\gamma \rightarrow \alpha + Fe_3C$

Use the Fe-Fe_3C phase diagram to calculate:

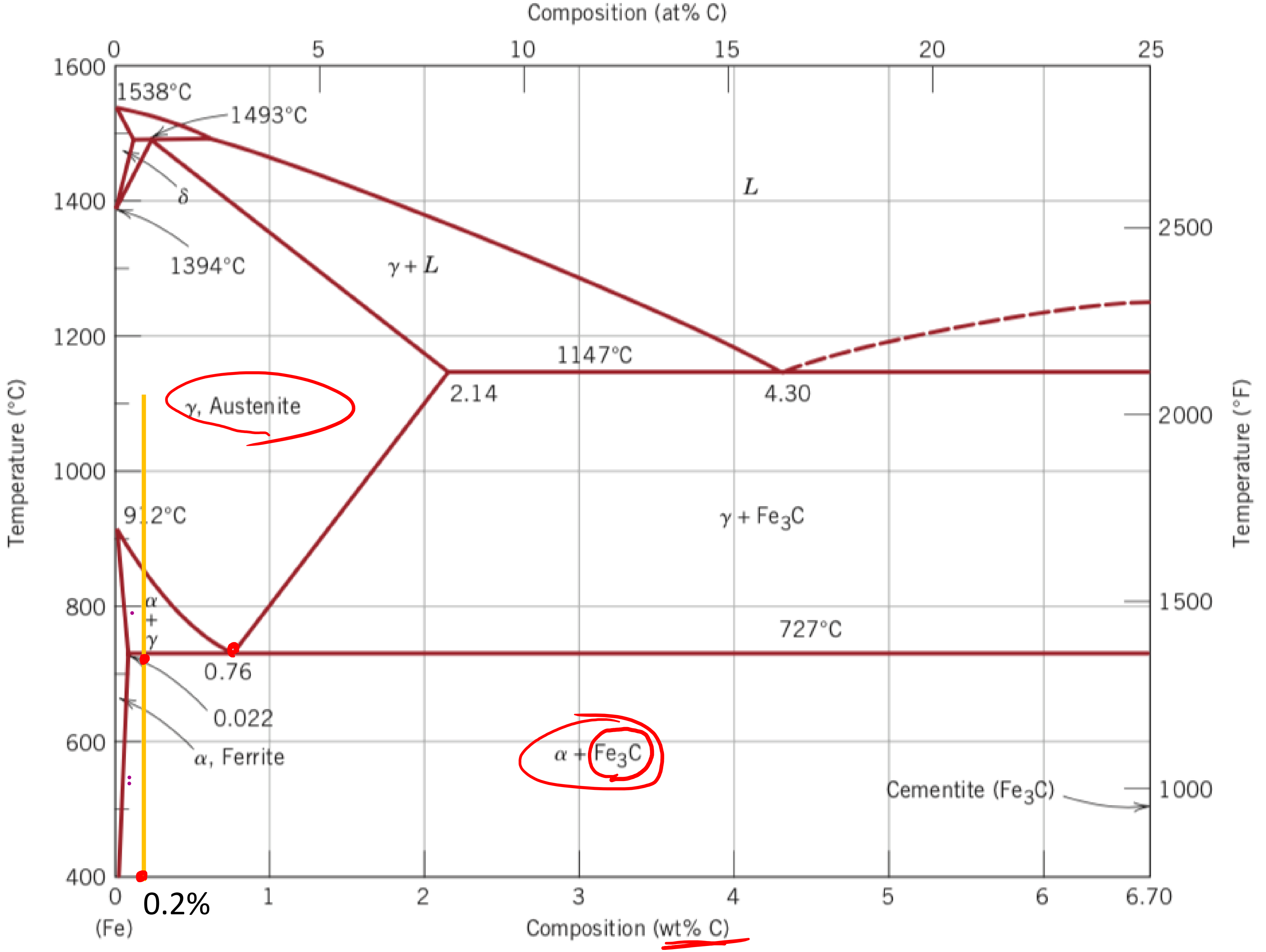
-the percentage of primary (pro-eutectoid) ferrite for a 0,2 % C steel

$$\alpha_p = \frac{0{,}76 - 0{,}2}{0{,}76 - 0{,}022} \cdot 100 =$$

$$= 76\%$$

PEARLITE = 24%

$Eut\,\alpha = ?$ $\quad$ ~~Eut~~ $(Fe_3C) = ?$



Use the Fe-Fe_3C phase diagram to calculate:

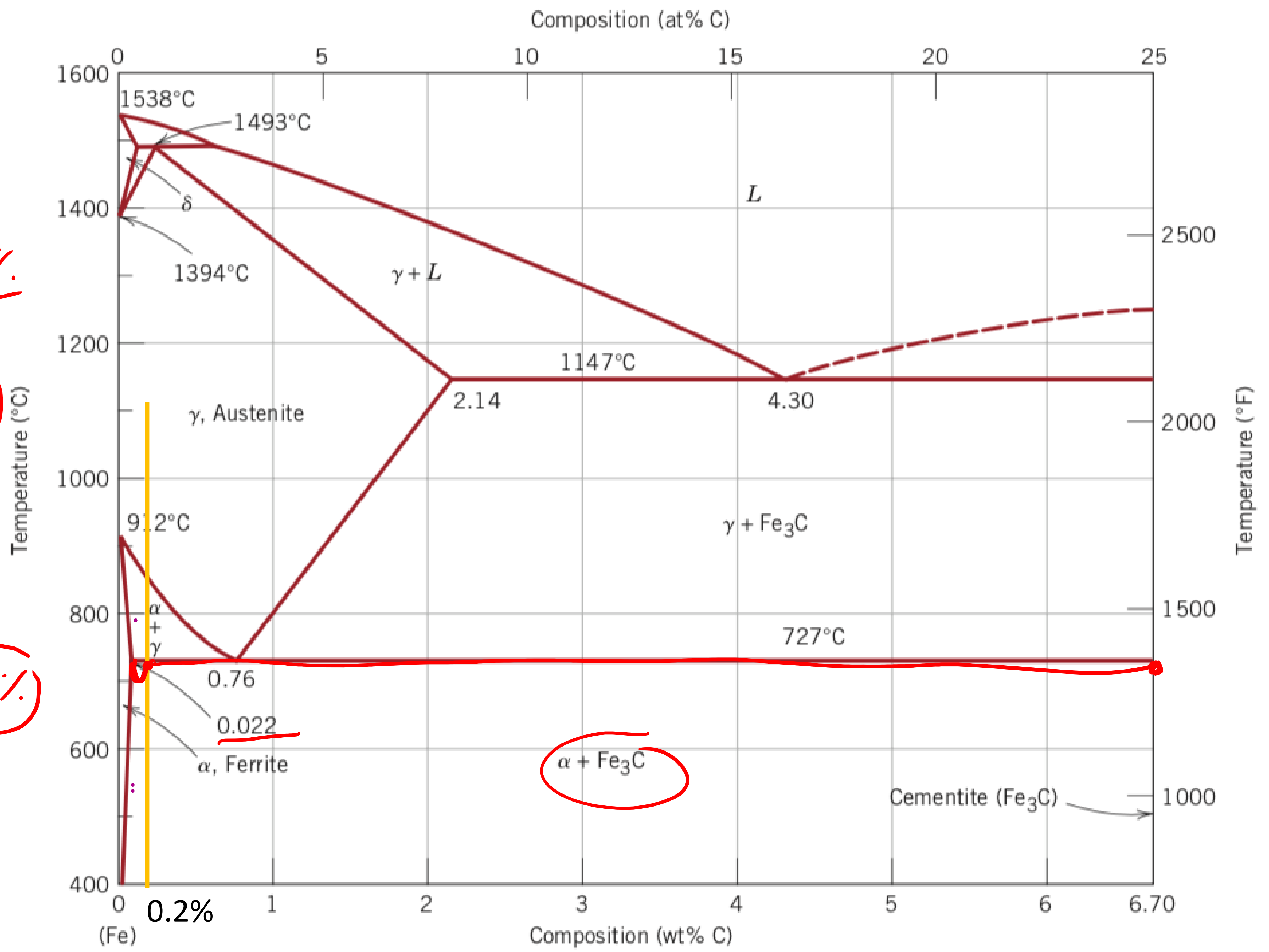
-the percentage of primary (pro-eutectoid) ferrite for a 0,2 % C steel

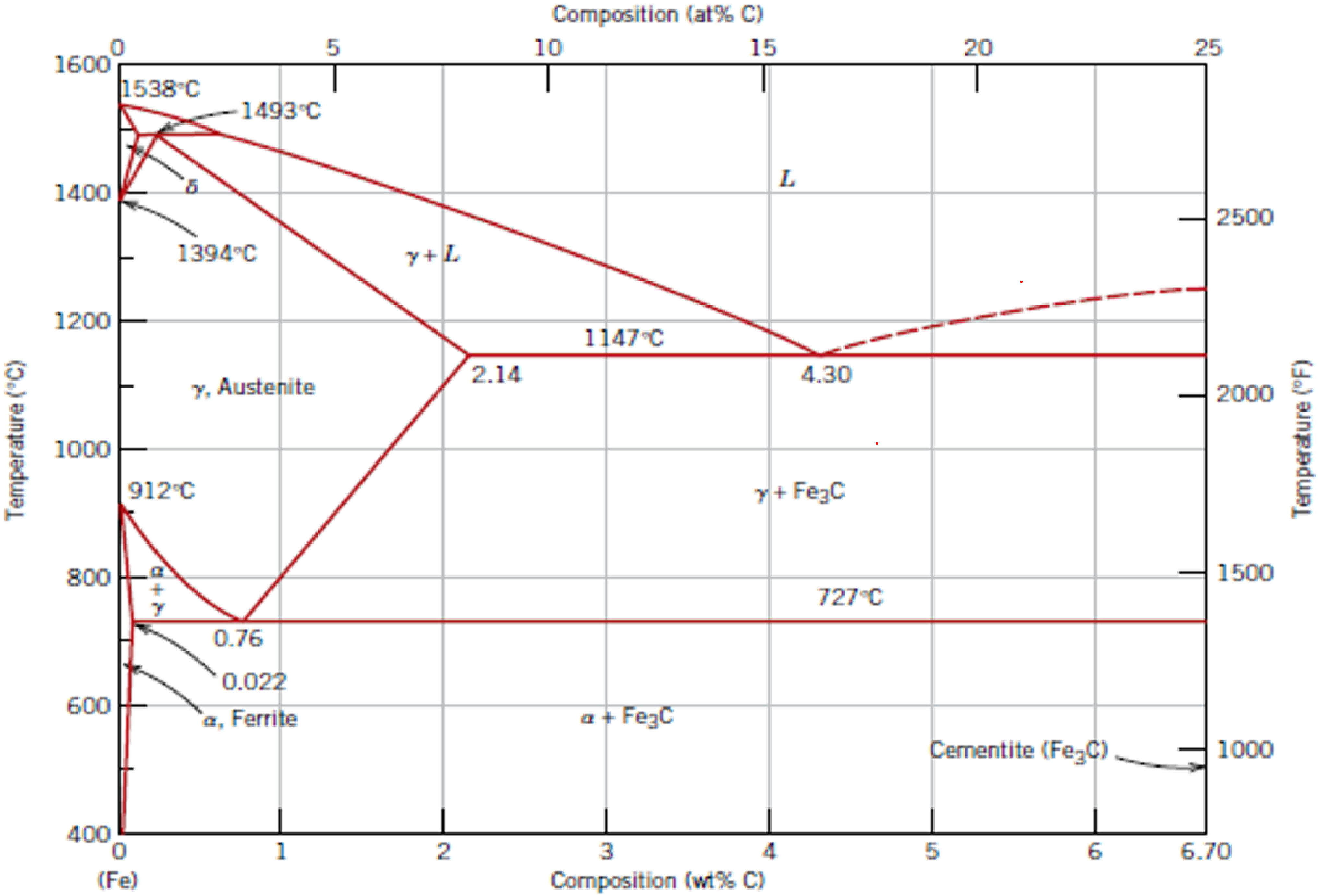
$\alpha_p = \underline{76\%}$ PEARL $= \underline{24\%}$

$$Fe_3C = \frac{0{,}2 - 0{,}022}{6{,}7 - 0{,}022} = 2{,}7\%$$

$$Eut\,\alpha = PEARL - Fe_3C =$$

$$= 24 - 2{,}7 = 21{,}3\%$$





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